

EXPERIMENT – 11

Performing IP Spoofing Using a GitHub Tool in Kali Linux (Educational Demonstration)

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Launch Terminal in Kali Linux.
2. Update system packages:
3. `sudo apt update`

Step 3: Clone IP Spoofing Tool from GitHub

1. Clone the repository:
2. `git clone https://github.com/<username>/<ip-spoofing-tool>`
3. Navigate to the tool directory:
4. `cd ip-spoofing-tool`

Step 4: Install Dependencies

1. Install required dependencies:
2. `sudo apt install python3`
3. `pip3 install -r requirements.txt`

Step 5: Configure the Tool

1. Open the script file:
2. `nano spoof.py`
3. Configure:
 - Target IP address
 - Spoofed source IP address
 - Network interface

Step 6: Execute IP Spoofing

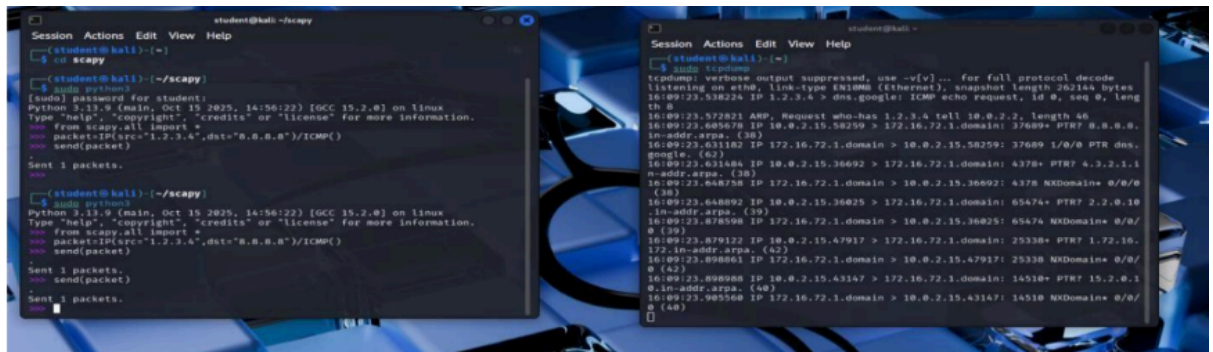
1. Run the tool with root privileges:
2. `sudo python3 spoof.py`
3. Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
2. `sudo tcpdump`

3. Verify spoofed source IP packets.

Output



The image displays two terminal windows from a Kali Linux system. The left window shows the execution of a scapy script that sends ICMP echo requests from source IP 1.2.3.4 to destination IP 8.8.8.8. The right window shows a tcpdump capture of the network traffic, displaying the raw packets and their decoded details, including the spoofed source IP.

```
student@kali: ~/scapy
[student@kali] (~)
$ scapy
[student@kali] (~)~/scapy
$ sudo python3
[sudo] password for student:
Python 3.11.9 (main, Oct 15 2025, 14:56:22) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> from scapy.all import *
>>> packet=IP(src="1.2.3.4",dst="8.8.8.8")/ICMP()
>>> send(packet)

Sent 1 packets.
>>>

[student@kali] (~)~/scapy
$ sudo python3
Python 3.11.9 (main, Oct 15 2025, 14:56:22) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> from scapy.all import *
>>> packet=IP(src="1.2.3.4",dst="8.8.8.8")/ICMP()
>>> send(packet)

Sent 1 packets.
>>>

[student@kali] (~)~/scapy
$ sudo tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:09:23.538224 IP 1.2.3.4 > dns.google: ICMP echo request, id 0, seq 0, leng
16:09:23.572821 ARP, Request who-has 1.2.3.4 tell 10.0.2.2, length 46
16:09:23.648578 IP 10.0.2.15.58259 > 172.16.72.1.domain: 37609+ PTR 8.8.8.8.
in-addr.arpa. (38)
16:09:23.651182 IP 172.16.72.1.domain > 10.0.2.15.58259: 37609 1/0/0 PTR dns.
google. (62)
16:09:23.631484 IP 10.0.2.15.36692 > 172.16.72.1.domain: 4378+ PTR 4.3.2.1.1
in-addr.arpa. (38)
16:09:23.648758 IP 172.16.72.1.domain > 10.0.2.15.36692: 4378 NXDomain+ 0/0/0
(38)
16:09:23.648892 IP 10.0.2.15.36025 > 172.16.72.1.domain: 65474+ PTR 2.2.0.10
in-addr.arpa. (39)
16:09:23.678506 IP 172.16.72.1.domain > 10.0.2.15.36025: 65474 NXDomain+ 0/0/
0 (39)
16:09:23.679122 IP 10.0.2.15.47917 > 172.16.72.1.domain: 25338+ PTR 1.72.16.
172.in-addr.arpa. (42)
16:09:23.698861 IP 172.16.72.1.domain > 10.0.2.15.47917: 25338 NXDomain+ 0/0/
0 (42)
16:09:23.698988 IP 10.0.2.15.43147 > 172.16.72.1.domain: 14510+ PTR 15.2.0.1
0.in-addr.arpa. (40)
16:09:23.905508 IP 172.16.72.1.domain > 10.0.2.15.43147: 14510 NXDomain+ 0/0/
0 (40)
[]
```

Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux. The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.